

SE LAB 3

System: Academic Elective Bidding & Allocation System

NAME: AADYA SANTHOSH

SRN: PES1UG24CS005

CLASS: 5A

Architecture Selection

We chose Layered Architecture for the Academic Elective Bidding & Allocation System.

Justification

- Separation of Concerns: It separates the front-end user interface from the complex seat allocation solver and schedule verifier algorithms.
- Modularity & Maintainability: Updates to the bidding logic or allocation engine can be made in the Business Layer without changing the UI or Database schema.

Security Advantage

Data Layer Isolation & Access Control: The Data Layer is strictly shielded from direct client connections.

All read and write requests must first be processed, authenticated, and authorized by the Business Layer. This tiering ensures that administrative override permissions (FR-005) are validated on the server side and that sensitive bidding data remains encrypted both in transit and at rest using AES-256 (NFR-002) before reaching the database.

Performance Benefit

Optimized In-Memory Processing & Low Latency: During peak bidding periods, the Business Layer can cache course capacities, schedules, and prerequisite rule sets locally in memory. By reducing round-trip SQL queries during bid collection and constraint validation, the Allocation Engine / Solver can execute batch seat assignments in under 30 seconds for 5,000 student bids, successfully fulfilling the NFR-001 performance requirements.